Hospital Patient Records – Pandas Project

Student Project Task Sheet

Project Objective

Analyze a large Hospital Patient Records dataset using Pandas. Complete the project using only the specified concepts.

Topics Allowed: head(), tail(), loc[], iloc[], Filtering, groupby(), Aggregation Functions, agg(), pivot_table()

Aggregation Functions: sum(), mean(), count(), min(), max(), median()

Dataset

File: Hospital_Patient_Records_12000.csv

Main columns: Patient_ID, Patient_Name, Age, Gender, Blood_Group, City, Department, Diagnosis, Admission_Type, Admission_Date, Discharge_Date, Length_of_Stay, Doctor_Fee, Room_Charge_Per_Day, Medicine_Cost, Lab_Cost, Room_Cost, Total_Bill, Insurance_Type, Insurance_Covered, Patient_Paid, Satisfaction_Score, Readmitted

Part 1 – head()

Task 1. Display the first 5 records.

Task 2. Display the first 10 records.

Task 3. Display the first 15 records.

Task 4. Display the first 20 records.

Part 2 – tail()

Task 5. Display the last 5 records.

Task 6. Display the last 10 records.

Task 7. Display the last 15 records.

Task 8. Display the last 20 records.

Part 3 – loc[]

Task 9. Display rows from index 0 to 9.

Task 10. Display rows from index 10 to 20.

Task 11. Using loc[], display rows 0 to 9 with Patient_Name, Age, Gender and Department.

Task 12. Using loc[], display rows 20 to 30 with Patient_Name, City, Diagnosis and Total_Bill.

Task 13. Using loc[], display rows 50 to 60 with Patient_ID, Age, Department, Insurance_Type and Patient_Paid.

Part 4 – iloc[]

Task 14. Display the first 5 rows using iloc[].

Task 15. Display rows from position 5 to 15.

Task 16. Display the first 10 rows and first 5 columns.

Task 17. Display rows 20 to 30 and columns 2 to 7.

Task 18. Display selected rows and columns using iloc[].

Part 5 – Filtering

Task 19. Find all patients whose Age is greater than 60.

Task 20. Find all patients whose Age is less than 18.

Task 21. Find all patients from Jaipur.

Task 22. Find all patients from the Cardiology department.

Task 23. Find all patients whose Total_Bill is greater than ■100,000.

Task 24. Find all patients whose Length_of_Stay is greater than 10 days.

Task 25. Find all patients whose Satisfaction_Score is less than 3.

Task 26. Find patients who are from Jaipur AND Cardiology department.

Task 27. Find patients who are from Delhi OR Mumbai.

Task 28. Find patients whose age is between 30 and 50.

Task 29. Find all patients with Emergency admission type.

Task 30. Find all patients diagnosed with Diabetes.

Task 31. Find patients where Age > 50 AND Total_Bill > ■75,000.

Part 6 – GroupBy + Aggregation

Task 32. Find the number of patients in each Department using groupby() + count().

Task 33. Find the total Total_Bill for each Department using sum().

Task 34. Find the average Total_Bill for each Department using mean().

Task 35. Find the minimum Total_Bill for each Department.

Task 36. Find the maximum Total_Bill for each Department.

Task 37. Find the median Total_Bill for each Department.

Task 38. Find the average Age of patients in each Department.

Task 39. Find the average Length_of_Stay for each Department.

Task 40. Find the total Medicine_Cost for each Department.

Task 41. Find the total Lab_Cost for each Department.

Task 42. Find the average Satisfaction_Score for each Department.

Part 7 – GroupBy Analysis

Task 43. Diagnosis: Find patient count, total billing and average billing for each Diagnosis.

Task 44. City: Find patient count, total billing and average billing for each City.

Task 45. Admission Type: Find patient count, average billing, minimum billing and maximum billing for each Admission_Type.

Task 46. Insurance: Find total Insurance_Covered and total Patient_Paid for each Insurance_Type.

Part 8 – Multiple Aggregations using agg()

Task 47. Department: calculate patient count, total billing, average billing, minimum billing and maximum billing.

Task 48. Diagnosis: calculate patient count, average Age, average Total_Bill and maximum Total_Bill.

Task 49. City: calculate patient count, total billing and average billing.

Task 50. Insurance_Type: calculate patient count, total billing, total Insurance_Covered and total Patient_Paid.

Task 51. Admission_Type: calculate patient count, average billing, minimum billing and maximum billing.

Part 9 – Pivot Table

Task 52. Create a pivot table showing total billing by Department and Gender.

Task 53. Create a pivot table showing average billing by Department and Gender.

Task 54. Create a pivot table showing patient count by Department and Admission_Type.

Task 55. Create a pivot table showing total billing by City and Department.

Task 56. Create a pivot table showing average Length_of_Stay by Department and Gender.

Task 57. Create a pivot table showing average Satisfaction_Score by Department and Gender.

Task 58. Create a pivot table showing total Insurance_Covered by Insurance_Type and Department.

Part 10 – Final Business Questions

Task 59. Which Department has the highest number of patients?

Task 60. Which Department has the highest total billing?

Task 61. Which Department has the highest average billing?

Task 62. Which Diagnosis has the highest number of patients?

Task 63. Which City generates the highest total billing?

Task 64. Which Admission_Type has the highest average billing?

Task 65. Which Insurance_Type has the highest total insurance coverage?

Task 66. Which Department has the highest average Satisfaction_Score?

Task 67. Which Department has the highest average Length_of_Stay?

Task 68. Which Department has the highest maximum Total_Bill?

Project Rules

Concept	Allowed
head()	Yes
tail()	Yes
loc[]	Yes
iloc[]	Yes
Filtering	Yes
groupby()	Yes
sum()	Yes
mean()	Yes
count()	Yes
min()	Yes
max()	Yes
median()	Yes
agg()	Yes
pivot_table()	Yes

Do Not Use: sort_values(), apply(), drop(), rename(), date operations, merge(), join(), or advanced Pandas concepts.

Submission Requirements

• Submit the Python/Jupyter Notebook file (.ipynb).

• Submit the Hospital CSV dataset.

• Provide proper output for every task.

• Add clear comments before each task.

• Write final answers for the Business Questions.

Recommended Notebook Structure

Import Pandas → Load Dataset → head() → tail() → loc[] → iloc[] → Filtering → GroupBy + Aggregation → agg() → Pivot Table → Final Business Questions

Goal: Build confidence in data selection, filtering, grouping, aggregation and summary analysis using Pandas.